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EXPERIMENT TITTLE: MOMENT OF INERTIA OF A RIGID OBJECT

Objectives :

1. Determine the torsional spring constant.
2. Determine the moment of inertia of a solid sphere, a disk, and a solid cylinder.
3. Determine and compare the theoretical moment of inertia with the experimental results.

SUMMARY EXPERIMENT

INTRODUCTION

Moment of inertia is a measure of the resistance an exhibits when a torque acts on it, causing a change in its rotational motion. Oscillatory and rotational motion in a system can be analyzed through a fundamental understanding of period and frequency. The period of periodic motion in a rigid in rotation is greatly influenced by the moment of inertia (I) and depends heavily on how the mass is distributed relative to its axis of rotation [1]. Mass distribution affects the dynamics of a system when an undergoes a combination of rotational motion about a fixed axis and back and forth motion across a point of equilibrium, known respectively as rotational and oscillatory motion [2]. For rotational oscillation to continue, the system requires a driving force in the form of angular momentum known as torque (τ). When the system is displaced from its initial position, the elasticity of the spiral wire triggers a restoring torque, whose direction is always opposite returning the system to its state of equilibrium. [3]. The magnitude of this motion is controlled by the spring stiffness, which is expressed by the spring constant or torsional constant kappa (κ). According to Hooke’s Law, the spring constant defines a linear relationship between the magnitude of the restoring torque and the angular displacement [4]. The interaction between the spiral constant, restoring torque, and moment of inertia is interrelated in determining the period of the system. Based on theory, the moments of inertia of a solid sphere, a solid cylinder, and a hollow cylinder rotating about their centers of mass are summarized in Table 1.

Tabel 1. Persamaan teoritis momen inersia benda tegar terhadap pusat massanya.

Rigid	Moment of Inertia
Solid Square	$\frac{2}{5}MR$
Solid Cylinder dan Disk	$\frac{1}{2}MR$
Hollow Cylinder	MR
Stem	$\frac{1}{12}MR$

The translational motion (F) of a load is expressed by the following equation mass m (kg) multiplied by acceleration a (m/s^2):

$$\Sigma F = ma \tag{1}$$

Next, the rotational motion (τ) of the wheel is expressed in Equation 2 as the moment of inertia (kg/m^2) multiplied by the rotational acceleration (m/s^2) as shown below,

$$\Sigma \tau = I\alpha \tag{2}$$

The moment of inertia of a rigid body in the experiment is obtained using the value of one period (T) in seconds in Equation 3 below:

$$I_{percobaan} = \kappa \left(\frac{T}{2\pi}\right)^2 \tag{3}$$

The kappa value (κ) is obtained using the following Equation 4:

$$\kappa = \frac{4(m_1+m_2)\pi^2}{m_{gradien}} \tag{4}$$

The kappa value (κ) is obtained by applying the mass values of loads 1 and 2, as shown in Figure A, in kilograms, by measuring the oscillation period of a transverse rod with equal masses attached to both ends to demonstrate that the moment of inertia is directly proportional to the square of the distance of the mass from its axis of rotation, and to determine the magnitude of the restoring torque from the spiral spring in the apparatus used. The gradient or slope value on the graph showing the relationship between period (T) and distance (r) in meters will be used in Equation 4 to calculate the kappa (κ) value in N·m.

METODOLOGY

A. Tools dan Materials

The following are the equipment and materials used in this lab:

Table 1. Tools and materials for the rigid body moment of inertia experiment.

Apparatus	Quantity	Function
Moment of inertia apparatus	1 set	Serves as the support and main system for the experiment
Ruler	1 piece	Used to measure the 's distance
Stopwatch	1 piece	Used to measure the 's oscillation time
Weight	1 piece	being measured
Solid cylinder	1 piece	being measured
Hollow cylinder	1 piece	being measured
Disk	1 piece	being measured
Solid sphere	1 piece	being measured
Rod	1 piece	being measured
Supporting plate	1 piece	being measured
Digital balance	1 piece	Used to measure the 's mass

B. Work Steps

The following are the procedural steps for the experiment on the moment of inertia of a rigid body shown in Figures A and B

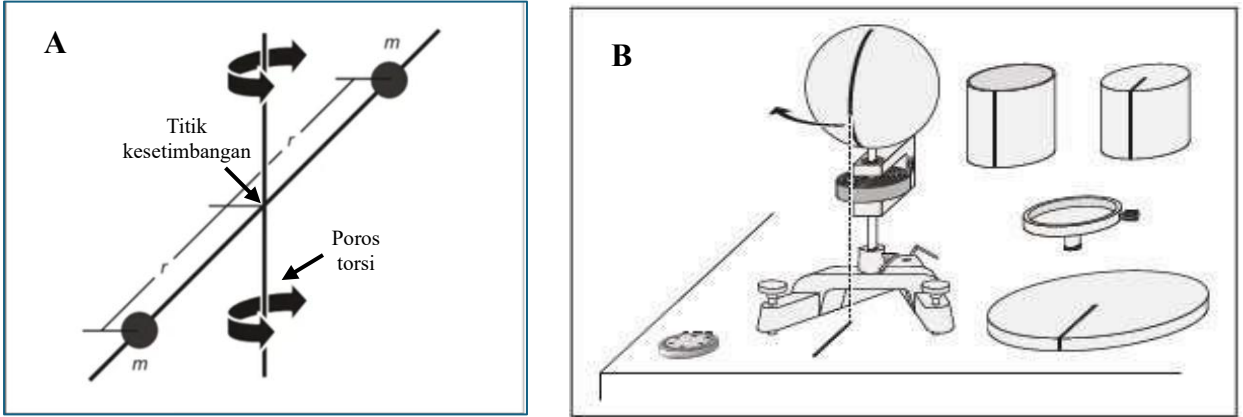


Figure A. Symmetric point-mass system; Figure B. Schematic of the experimental setup for the moment of inertia of a rigid body [5],[6]

This experiment was conducted in two stages: determining the kappa (κ) value and subsequently determining the moment of inertia for s of different shapes.

- A. The first part determines the value of kappa (κ)
1. Measure the mass of the load (m) used in the experiment to determine the value of kappa.
 2. Set up the apparatus as shown in Figure A
 3. Determine the distance between the mass and the equilibrium point (r)
 4. Rotate it through an angle of 180° and then release it.
 5. Record the time required for 5 oscillations in the experiment table.
 6. Perform the same steps for different distances between the mass and the equilibrium point
- B. The second part determines the moment of inertia of the s.
1. Set up the apparatus as shown in Figure B.
 2. Calculate the mass and radius of each .
 3. Position the sphere on the torsion axis, then mark its equilibrium position.
 4. Rotate the sphere by 180° and then release it.
 5. Record the time required for 5 oscillations in the experiment table.
 6. Calculate the period (T) from the 5 oscillations.
 7. Replace the sphere with other s (disk, supporting plate, solid cylinder).
 8. Repeat steps 2–6.
 9. Perform 5 trials for each experiment.

D. Flowchart

DATA ANALYSIS AND DISCUSSION

A. Analysis of Measurement Data

The following is a table of data from measurements conducted in the laboratory:

Table 1. Results of kappa (κ) value measurements.

Repitition	Time for 5 Oscillations (seconds)						
	r = 0 m	r = 0,05 m	r = 0,10 m	r = 0,15 m	r = 0,20 m	r = 0,25 m	r = 0,30 m
1							
2							
3							
4							
5							
Average							
Periode							

Table 2. Measurements of the moments of inertia of rigid bodies

Rigid Body	m (kg)	R (m)	Time Taken for 5 Oscillations in the —th Experiment					Average	T (s)
			1	2	3	4	5		
Plate									
Solid Cylinder + Plate									
Solid Cylinder									
Solid Sphare									
Disk									

*coloured block: this does not need to be filled in, as the theoretical values of the moments of inertia of the plate and the solid cylinder are not required, and the experimentally determined moment of inertia of the solid cylinder is obtained by subtracting the moment of inertia of the plate from the sum of the moments of inertia of the solid cylinder and the plate

Approved by:

Laboratory Assistant

B. Graph

Graph the relationship between the load distance (r) and the period (T) of the oscillating system to determine the following kappa values.

(NOTE: create a graph of the x-axis for the r^2 value and the y-axis for the T^2 value)

C. Calculations

Example calculation

(NOTE: Determine the kappa value and moment of inertia for each variation object)

The following is a table of the calculation results obtained:

Table 3. Results of moment of inertia calculations for a rigid body

Rigid	Slope/Gradient Value	Kappa (N.m)	$I_p(kg/m^2)$	$I_t(kg/m^2)$	Error (%)
Silinder Pejal					
Bola Pejal					
Disk					

Note: I_p (calculated moment of inertia), I_t (theoretical moment of inertia)

The error (%) is the percentage of the calculated value relative to the theoretical value
The following are the calculated values of the moments of inertia for a different s in this experiment

D. Discussion

CONCLUSIONS

From this practical session, it can be concluded that:

- 1.
- 2.
- 3.

REFERENCES

- [1] R. A. Serway and J. W. Jewett, *Physics for Scientists and Engineers with Modern Physics*, 9th ed. Boston, MA: Brooks/Cole Cengage Learning, 2014.
- [2] P. A. Tipler and G. Mosca, *Physics for Scientists and Engineers*, 6th ed. New York, NY: W. H. Freeman and Company, 2008.
- [3] D. Halliday, R. Resnick, and J. Walker, *Fundamentals of Physics*, 10th ed. Hoboken, NJ: John Wiley & Sons, 2014.
- [4] R. A. Serway and J. W. Jewett, *Physics for Scientists and Engineers with Modern Physics*, 9th ed. Boston, MA: Brooks/Cole Cengage Learning, 2014
- [5] “Mechanics 0314-Sel/Kem.”
- [6] “Mechanics 0314-Sel/Kem.”

PRELIMINARY ASSIGNMENT

1. Explain what is meant by a rigid !
2. Explain the relationship with Newton's second law of motion in this experiment!
3. Explain the value of kappa (κ) and explain how to calculate the value of kappa mathematically! (This can be found in the ITS Basic Physics Laboratory YouTube video).
4. Derive the formula for the moment of inertia given in Table 1 (Introduction) and derive Equation 3!
5. Explain what factors influence the moment of inertia of a body!
6. Explain restoring torque!